

INNOASIC IP LIST- Q1 2007 - confidential

Product Code	Product Name	Deliverable		Description
		Fixed	Configurable	
BA1210	32bit RISC Processor	available	available	32bit embedded processor in the same class with ARM9 as processors
DSP-010	Jazz 2020 Processor	available	available	Dual-MAC DSP Processor
DSP-011	Jazz 2040 Processor	available	available	Quad-MAC DSP Processor
JSK-VOP-010	Jazz Voice VPP	available	available	Full voice processing and verification environments for home VoIP solutions including VoIP framework, G.7xx codecs package, G.168 LEC
JSK-VOP-011	Jazz Voice VID	available	available	Same as above with video codec.
JSK-VOP-020	Jazz Voice IAD	available	available	Full voice processing and verification environments for high channel IAD device including VoIP framework, G.7xx codecs package, G.168 LEC
JSK-MED-010	Jazz Media AAP	available	available	Audio decoding and encoding solutions. including MPEG L1/L2, MP3, AAC/AAC+, BSAC, WMA, Dolby Digital
JSK-MED-020	Jazz Media MMP	available	available	CIF/QVGA video decoding and encoding solutions including video, audio, image codecs. including H.264 AVC, MPEG4, WMV, MPEG L1/L2, MP3, AAC/AAC+, BSAC, WMA, JPEG
JSK-MED-021	Jazz Media MMP w/ speech codecs	available	available	CIF/QVGA video decoding and encoding solutions including video, audio, image codecs. Sames as above with audio and speech codecs ; G.7xx codecs package
JSK-MED-030	Jazz Media BMP	available	available	D1(NTSC/PAL) video decoding and encoding solutions including video, audio, image codecs. including H.264 AVC, MPEG4, WMV, MPEG L1/L2, MP3, AAC/AAC+, BSAC, WMA, JPEG
JSK-MED-031	Jazz Media BMP w/ speech codecs	available	available	D1(NTSC/PAL) video decoding and encoding solutions including video, audio, image codecs. Sames as above with audio and speech codecs ; G.7xx codecs package
JSK-MED-040	Jazz Media HD	available	available	High definition video decoding and encoding solutions including video, audio, image codecs. including H.264 AVC, MPEG4, WMV, MPEG L1/L2, MP3, AAC/AAC+, BSAC, WMA, JPEG
JRB-RH1-020	Jazz PSA Rehearsal F2	available	available	PCI board for emulating Jazz DSP and other logic; populated with dual Virtex4 FPGAs
JSW-STD	Jazz Standard Tool Suite	available	available	Additional copies of the tool suite for software development on Jazz DSPs
JSW-CMP	Jazz Composer Tool Suite	available	available	Additional copies of the tool suite for processor/core configuration and software development on Jazz DSPs
OPT-010	FPU DDCU	available	available	Supporting floating point unit DDCU into a Jazz Platform
OPT-011	CCIR-656 Codec			Supporting CCIR-656 codec into a Jazz Platform
OPT-012	TV Encoder			Supporting NTSC/PAL Encoder into a Jazz Platform
OPT-013	Scaleable Video Codec			Supporting Scaleable Video Coding into a Jazz Platform
		FPGA Netlist	RTL	
H264C	H.264/MPEG4 Main/High/Baseline Video Codecs	-	available	H.264/MPEG4 Main/High/Baseline Video Encoder and Decoder
H261C	H.261/3 Video Codecs	-	available	H.261/3 Video Encoding and Decoding
		FPGA Netlist	RTL	
CWda01	SPDIF-AES/EBU Feed Forward-Receiver	-	available	
CWda02	SPDIF-AES/EBU Feed Forward-Transmitter	-	available	
CWda03	SPDIF-AES/EBU to I2S Converter	-	available	
CWda04	I2S to SPDIF-AES/EBU Converter	-	available	
CWda06	I2S Transmitter	-	available	

CWda07	SPDIF-AES/EBU Self-Clocked Receiver	-	available	
CWda10	I2S Receiver	-	available	
CWda30	Stereo Digital Audio Sigma-Delta Modulator	-	available	
CWda41	SDI Audio De-Embedder	-	available	
CWda50	Low Cost Audio Sample Rate Converter	-	available	
CWda52	Multi-Channel Audio Sample Rate Converter	-	available	
		FPGA Netlist	RTL	
EC220	32bit PCI Master/Target	available	available	
EC125	32bit PCI Target	available	available	
EC140	64bit PCI Target	available	available	
EC150	PCI-ISA Bridge	available	available	
EC300	PCI Arbiter	available	available	
EC310	PCI Express End Point	available	available	
EP430	32bit PCI Host Bridge	available	available	
EP420	64bit PCI Host Bridge	available	available	
EP440	PCI-PCI Bridge	available	available	
EP423	PCI-X Host Bridge	-	available	
EP243	PCI-X Master/Target	-	available	
EP201	PowerPC Master	available	available	
EP100	PowerPC Slave	available	available	
EP433	PowerPC-PCI Host Bridge	available	available	
EP300	PowerPC Arbiter	available	available	
EP505	EC-SDR SDRAM Controller	available	available	
EP455	EC-PCI Host Bridge	available	available	
EP140	AHB Slave	available	available	
EP240	AHB Master	available	available	
EP505	AHB-SDR SDRAM Controller	available	available	
EP534	AHB-DDR SDRAM Controller	available	available	
EP435	AHB-PCI Host Bridge	available	available	
EP246	AHB-DMA Controller	available	available	
EP150	MIPS SysAD Slave	available	available	
EP155	MIPS EC Slave	available	available	
EP450	SysAD-PCI Host Bridge	available	available	
EP500	Flash/EEPROM/SRAM Controller	available	available	
EP501	NAND Flash Controller	available	available	
EP510	CompactFlash Host Controller	available	available	
EP512	CompactFlash with EXCA Registers	available	available	
EP513	CompactFlash Slave Controller	available	available	
EP525	SDR SDRAM Controller	available	available	
EP530	DDR SDRAM Controller	available	available	
EP532	DDR2 SDRAM Controller	available	available	
EP550	SD/SDIO Host Controller	available	available	
EP560	SD/SDIO Slave Controller	available	available	

EP600	UART	available	available	
EP660	DMA Controller	available	available	
EP610	Interrupt Controller	available	available	
EP620	Timer	available	available	
ES100	PowerPC System Controllers	-	available	
ES400	AHB System Controllers	-	available	
ES500	SysAD System Controllers	-	available	
ES600	AMBA Peripheral Package	-	available	
		FPGA Netlist	RTL	
SDVBH	DVB-H Demodulator	-	available	
SAMIM	AMBA Multi-Layered	-	available	
SDMAC	DMA Controller	-	available	
SAVIC	Advanced Vectored Interrupt Controller	-	available	
SXMI	External Memory Controller	-	available	
SDRAMC	Dynamic RAM Memory Controller	-	available	
SSCI	Smart Card Controller	-	available	
SATAH	ATA Host Controller	-	available	
SLCDC	LCD Controller	-	available	
SDES	DES & Triple DES Co-Processor	-	available	
SRSAC	RSA Co-Processor	-	available	
		FPGA Netlist	RTL	
AADMA	AHB-DMA Controller	-	available	
AASPI	AHB-SPI Controller	-	available	
AAIC	APB-I2C Master/Slave Controller	-	available	
AACF	AHB-Compactflash Controller	-	available	
AAPCI	AHB-PCI CardBus Controller	-	available	
ANANDF	NAND Flash Controller	-	available	
AUART	UART	-	available	
ASDIOH	SD/SDIO Host Controller	-	available	
ASDIOD	SD/SDIO Device Controller	-	available	
ACEATAM	CE-ATA/MMC Controller	-	available	
AUSBH11	USB1.1 Host	-	available	
AUSBD11	USB1.1 Device	-	available	
AUSBHU11	USB1.1 Hub	-	available	
AUSBH20	USB2.0 Host	-	available	
AUSBD20	USB2.0 Device	-	available	
AUSBHU20	USB2.0 Hub	-	available	
AUSBOTG20	USB2.0 OTG	-	available	
		FPGA Netlist	RTL	
GPEX-EP	PCI Express End Point	-	available	
GPEX-RC	PCI Express Root Complex	-	available	

GPEX-SW	PCI Express Switch Controller	-	available				
GPEX-AS	Advanced Switing Controller	-	available				
GPAB	PCIe-AXI Bridge	-	available				
GHT	Hyper Transport Controller	-	available				
GRIO	Rapid IO Controller	-	available				
GEXMAC	10 Gigabit Ethernet MAC Controller	-	available				
GEMAC	10/100/1000 Ethernet MAC Controller	-	available				
GSPI42-MC	SPI4.2 Controller	-	available				
GPUL3	POS-UTOPIA Level 3	-	available				
GA97	AC97 Controller	-	available				
GUSB20D	USB2.0 Device Controller	-	available				
GSCI	Smart Card Interface Controller	-	available				
GI2S	I2S Controller	-	available				
GFFTI	FFT/IFFT Controller	-	available				
GSDRAM	SDRAM Controller	-	available				
GDMA	DMA Controller	-	available				
GVED	Viterbi Encoder/Decoder	-	available				
GRSED	Reed Solomon Encoder/Decoder	-	available				
GUART	UART	-	available				
GHDLC	HDLC	-	available				
GT1E1F	T1/E1 Framer	-	available				
		FPGA Netlist	RTL				
MT100MAC	10/100 MAC	available	available				
MT1GMAC	10/100/1000 MAC	available	available				
MT1GMACA	10/100/1000 MAC with Acc	available	available				
MT1GXP	1000 Base-X PCS (8b/10b)	available	available				
MTPTG	PCIe to GMII	available	available				
MT2.5GMAC	1G/2.5G/10G MAC	available	available				
MT2.5GXP	2500 Base-X PCS (8b/10b)	available	available				
MT10GMAC	10G MAC	available	available				
MT10GHMAC	10G HiGig/HiGig2 MAC	available	available				
MT10RP	10G Base-R PCS (64/66)	available	available				
MT10GWP	10G Base-W PCS (64/66+WIS)	available	available				
MT10GXP	10G Base-X PCS (XAUI)	available	available				
MT10GBP	1G/10G Backplane	available	available				
MTSTX	SPI4.2 to XAUI Bridge	available	available				
MTL2ES	L2 Ethernet Switch	available	available				
MTL2ESQ	L2 Ethernet Switch with QoS	available	available				
MYL2ES1588	L2 Ethernet Switch with 1588 support	available	available				
	TSMC Process	0.35um	0.18um	0.13um	90nm		
ADC	8Bit 40Msps ADC	available	available				
	8Bit 44Msps ADC	available	available				

	8Bit 135MHz ADC		available			
	8Bit 205MHz ADC		available			
	10Bit 1usps ADC		available			
	10Bit 1Msps ADC	available	available			
	10Bit 10Msps ADC			available		
	10Bit 20Msps ADC	available	available	available		
	10Bit 27Msps ADC		available	available		
	10Bit 40Msps ADC	available	available	available		
	10Bit 44Msps ADC		available	available		
	10Bit 50Msps ADC	available	available			
	10Bit 88Msps ADC		available	available		
	10Bit 105MHz ADC		available			
	10Bit 120MHz ADC		available			
	10Bit 140MHz ADC		available			
	10Bit 165MHz ADC			available		
	12Bit 27Msps ADC			available	available	
	12Bit 54Msps ADC	available	available	available		
	12Bit 60Msps ADC	available	available			
	13Bit 80Msps ADC		available	available		
	24Bit 96KHz Audio ADC	available	available	available		
DAC	6Bit 88MHz DAC		available			
	8Bit 44Msps DAC	available	available			
	8Bit 88MHz DAC		available			
	9Bit 40Msps DAC		available			
	10Bit 200MHz RGB Video DAC		available	available		
	10Bit 250MHz RGB Video DAC			available		
	10Bit 80Msps DAC		available			
	10Bit 88Msps DAC		available			
	10Bit 160Msps DAC	available	available	available		
	10Bit 400Msps DAC	available	available			
	12Bit 200Msps DAC	available	available			
	13Bit 80Msps DAC		available			
	13Bit 160Msps DAC		available			
	16Bit 200Ksps DAC	available				
	24Bit 96KHz Audio DAC	available	available	available		
PLL	50MHz to 100MHz PLL			available		
	54MHz to 108MHz PLL	available	available	available		
	80MHz to 160MHz PLL		available	available		
	60MHz to 200MHz PLL		available	available		
	27MHz to 246MHz PLL	available	available	available		
	480MHz PLL		available			
	2.5G PLL		available			
	12MHz to 60Mhz input 133MHz output Jitter PLL		available	available		
	12MHz to 60Mhz input 266MHz output Jitter PLL		available	available		

	5MHz to 33Mhz input 700MHz output Jitter PLL		available	available			
CDR	480MHz CDR		available				
	2.5G CDR		available				
PHY	2.5G SerDes		available				
	PCI Express PHY		available				
	HDMI Transmitter/Receiver PHY		available				
DLL	240MHz DLL		available				
	480MHz DLL		available				
USB	USB1.1 Device PHY		available				
	USB1.1 OTG PHY						
	USB2.0 Device PHY		available				
	USB2.0 OTG PHY						
RF/IF	IF 1.24G AFE IF PGA		available				
	IF 1.24G AFE RF Mixer		available				
	IF 1.24G Baseband Gm-C Channel Filter		available				
	IF 1.24G Baseband AGC		available				
	IF 1.24G PLL		available				
	RF 2.5G AFE IF PGA		available				
	RF 2.5G AFE RF Mixer		available				
	RF 2.5G PLL		available				
Converter	DC-DC Converter		available	available			
LDO	Voltage Regulator		available	available			
Amplifier	Headphone Amplifier		available				
	Audio Mixer, Volume Control, Mic-phone Bias		available				
	Gain Amplifier	available	available				
	Audio Speaker Amplifier	available	available				
	Low Noise Amplifier(LNA)		available				
Oscillator	32KHz 1uA Crystal Oscillator		available				
	50MHz 0.5mA Crystal Oscillator	available	available				